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A Mini-Project Report
On

“GAS LEAK DETECTION WITH ALERT”

Submitted in partial fulfillment of the requirements as a part of the curriculum,

Bachelors of Engineering in Artificial Intelligence and Machine Learning

Submitted by

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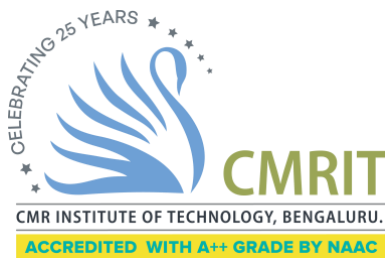


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2025-2026**

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CERTIFICATE

Certified that the Mini Project entitled “**GAS LEAK DETECTION WITH ALERT**” is presented by **Ms. PUNYASHREE L P & Ms. PAVITHRA N**, bonafide student of **CMR Institute of Technology** in partial fulfillment of the requirements as a part of the curriculum, **Bachelors of Engineering in Artificial Intelligence and Machine Learning**, of **Visvesvaraya Technological University, Belagavi** during the year **2025-26**. It is certified that all correction/suggestion indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Mini Project report has been approved as it satisfies the academic requirements in respect of the Mini Project prescribed for the bachelor of engineering degree.

Ms. VINUTA S

Signature of the Guide

(Dr. Shyam P Joy)

Signature of the HOD

GAS LEAK DETECTION WITH ALERT

ABSTRACT

Gas leakage poses a significant threat to safety in domestic, commercial, and industrial environments. Early detection of gas leakage is essential to prevent fire hazards, explosions, and health risks. This mini-project presents the design and implementation of a Gas Leakage Detection System using an Arduino microcontroller and an MQ-2 gas sensor to continuously monitor the surrounding air for the presence of combustible gases.

The system incorporates a calibration mechanism to determine the normal air reference value, thereby minimizing false detections during initial operation. Based on the measured gas concentration, the system categorizes the condition into normal, slight gas leakage, and major gas leakage levels using predefined threshold values. Visual and audible alerts are provided through an LED and a buzzer, which are activated only when the gas concentration exceeds the safe limit.

The proposed system is simple, economical, and reliable, making it suitable for real-time gas monitoring and safety applications. This project demonstrates an effective hardware–software integration approach for enhancing gas leak detection accuracy and improving overall safety.

ACKNOWLEDGEMENT

It is my proud privilege and duty to acknowledge the kind of help and guidance received from several people in preparation of this report. Apart from my own, the success of this report depends largely on the encouragement and guidelines of many others. It would have not been possible to prepare this report in this form without their valuable help, co- operation and guidance.

I would like to express my deep sense of gratitude to our Principal **Dr. Sanjay Jain**, CMR Institute of Technology College, Bangalore for his motivation and for creating an inspiring atmosphere in college by providing state of the art facilities for preparation and delivery of this report.

My sincere thanks to **Dr. Shyam P Joy**, Head of Department of Artificial Intelligence and Machine Learning, for his whole-hearted support in completion of the Mini Project.

I am highly indebted to my guide Assistant Prof .**Ms. VINUTA S**, for guiding and giving timely advices and suggestions in the successful completion of this seminar.

My sincere thanks to Assistant Prof. **Ms. Sandhya Kumari**, mini project coordinator for having supported the work related to this project.

Last but not least, I would like to put forward my heartfelt acknowledgement to all my classmates, friends and all those who have directly or indirectly provided their overwhelming support during my seminar work and the development of this report.

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CHAPTER 1

INTRODUCTION

1.1 General Introduction

Safety has become a critical concern in today's rapidly growing domestic, commercial, and industrial environments. With the widespread use of combustible gases such as Liquefied Petroleum Gas (LPG), propane, methane, and butane for cooking, heating, and industrial operations, the risk of accidental gas leakage has increased significantly. Even a small undetected gas leak can lead to severe consequences such as fire hazards, explosions, environmental pollution, property damage, and loss of human life. Therefore, early detection and timely alerting of gas leakage is essential to ensure safety and prevent disasters.

Traditional gas leak detection methods often rely on manual inspection or basic alarm systems that either lack accuracy or fail to differentiate between minor and major leak conditions. In many cases, such systems generate false alarms due to environmental variations or sensor noise, which reduces their reliability and user trust. Moreover, most low-cost systems do not provide graded alerts, making it difficult for users to assess the severity of the leak and respond appropriately.

To overcome these limitations, embedded system-based safety solutions have gained significant importance. Microcontroller-based gas detection systems provide an efficient, cost-effective, and reliable approach for continuous monitoring of gas concentration in the surrounding environment. By integrating gas sensors with microcontrollers, real-time monitoring, data processing, and alert generation can be achieved effectively.

This project titled “**Gas Leak Detection and Alert System Using Arduino**” focuses on the design and implementation of a smart gas monitoring system capable of detecting gas leakage levels and categorizing them into **normal, slight leak, and high leak conditions**. The system uses an analog gas sensor (MQ-series) interfaced with an Arduino microcontroller to continuously measure gas

concentration and compare it with a calibrated baseline air reference value. Based on the percentage deviation from normal air conditions, appropriate visual and audible alerts are generated using an LED and a buzzer.

The proposed system is designed to be simple, low-cost, reliable, and suitable for deployment in households, kitchens, laboratories, gas storage areas, and small-scale industries. By providing early warnings and graded alerts, the system enhances safety and helps prevent catastrophic accidents caused by gas leakage.

1.2 Project Motivation and Objectives

Gas leakage is a major safety concern in homes and workplaces, as delayed detection can lead to fire, explosion, and loss of life. In many cases, gas leaks remain unnoticed until the concentration reaches a dangerous level. This creates a strong need for a system that can continuously monitor gas levels and provide early warnings to prevent accidents.

Conventional gas detectors usually work on a simple ON/OFF mechanism and do not indicate the severity of leakage. This may cause either unnecessary panic or insufficient warning. To overcome this limitation, the proposed system introduces a multi-level alert mechanism that classifies gas leakage as normal, slight, or high based on the percentage deviation from a calibrated baseline air value. Sensor averaging is used to improve accuracy and reduce false alarms. Distinct LED and buzzer indications are provided for each condition to clearly convey the severity of the leak.

Objectives of the Project

- To design and implement a gas leak detection system using an Arduino microcontroller.
- To calibrate the sensor and establish a baseline air reference value.
- To classify gas leakage into normal, slight, and high levels.
- To provide appropriate visual and audible alerts based on leak severity.
- To develop a low-cost, user-friendly safety system for domestic and industrial applications.

CHAPTER 2

DESIGN AND IMPLEMENTATION

2.1 System Design

The Gas Leak Detection System is designed to monitor the presence of combustible gases in the environment and alert users in case of a leak. The system integrates an Arduino microcontroller, a gas sensor module, an LED, and a buzzer to provide real-time detection and warning.

The system is built for simplicity, reliability, and real-time response. Its core principle is to measure the gas concentration and compare it with a calibrated baseline. Depending on the deviation, the system activates visual and audio alerts.

The system consists of four main components:

1. Gas Sensor Module (MQ Series)

- Detects gas concentration and outputs an analog signal.

2. Arduino UNO

- Reads sensor data, performs calculations, and triggers alerts.

3. Alert System

- **LED:** Visual warning indicator.
- **Buzzer:** Audio alarm for leak detection.

4. Power Supply

- Provides regulated 5V supply to Arduino and peripherals.

Working principle

The sensor measures gas in the air. The Arduino compares the sensor reading with a reference value obtained during calibration. Alerts are activated if the deviation crosses defined thresholds.

Hardware Components: Aurdino UNO, Gas sensor, LED, Buzzer, Resistor, Jumper wires, Breadboard

2.2 System Implementation

Software Implementation: Arduino IDE (C/C++) is used to:

1. Calibrate sensor by averaging readings in clean air.
2. Read multiple sensor values to reduce noise.
3. Compute percentage deviation from baseline.
4. Activate LED and buzzer alerts based on thresholds.

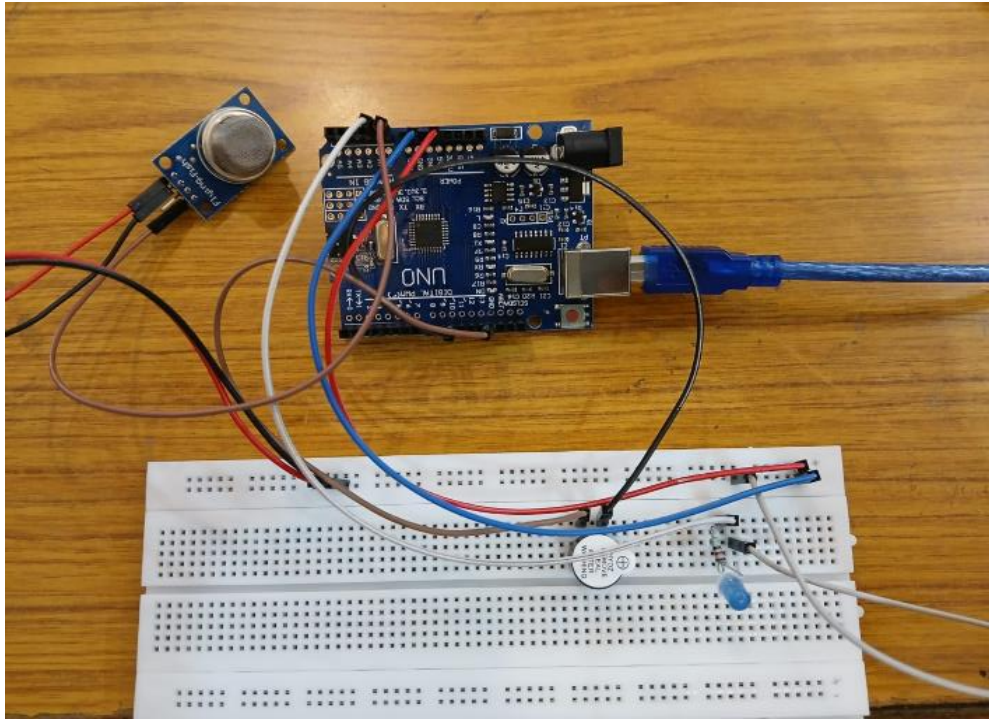
Calibration: On startup, the sensor takes 50 readings at 1-second intervals. The average forms the baseline (air_ref) for detecting leaks.

Detection Algorithm:

- **Normal Air:** Deviation $< 10\%$ $\rightarrow$ LED & Buzzer OFF
- **Slight Leak:** Deviation $10\text{--}25\%$ $\rightarrow$ LED blinks, buzzer beeps
- **Major Leak:** Deviation $> 25\%$ $\rightarrow$ LED ON, buzzer continuous

Serial Monitoring: Displays sensor value, percentage deviation, and status for debugging and testing.

Outcome: The system successfully detects and differentiates gas leaks, providing reliable real-time alerts. It is cost-effective and suitable for domestic and industrial safety



2.1. HARDWARE SETUP

CHAPTER 3

TESTING / RESULT AND ANALYSIS

Testing Methodology

The system was tested to verify accuracy, alert mechanism, and responsiveness:

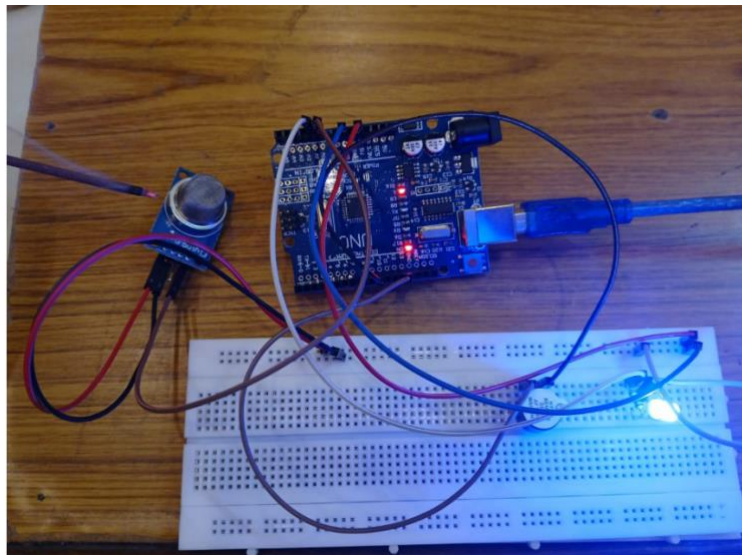
1. Calibration Test:
 - Performed in clean air; 50 readings averaged to get air reference value
 - Baseline was stable, ensuring reliable detection.
2. Slight Gas Leak Test:
 - Small gas concentration introduced.
 - LED blinked and buzzer beeped for 10–25% deviation.
3. Major Gas Leak Test:
 - Higher gas concentration applied.
 - LED stayed ON and buzzer sounded continuously for >25% deviation.

RESULT:

Condition	Sensor Deviation	LED	Buzzer	Serial Monitor
Normal Air	<10%	OFF	OFF	"Status: Air Normal"
Slight Leak	10–25%	Blink	Short Beep	"Slight Gas Leak Detected"
Major Leak	>25%	ON	Continuous	"Gas Leak Detected! (HIGH LEVEL)"

Analysis:

- Calibration provided a stable baseline for normal air.
- Threshold-based detection correctly distinguished slight and major leaks.
- Averaging readings reduced noise and prevented false alarms.
- Alerts (LED & buzzer) were immediate and accurate.
- Extremely low gas concentrations ($<10\%$) were not detected, which is acceptable for safety thresholds.



3.WORKING MODEL OF GAS LEAK DETCTION WITH ALERT

CHAPTER 4

CONCLUSION AND FUTURE ENCHANCEMENTS

Conclusion

The Gas Leak Detection System successfully monitors combustible gas levels and provides real-time alerts using LED and buzzer. Calibration and threshold-based detection ensure accurate differentiation between normal air, slight leaks, and major leaks. The system is reliable, cost-effective, and suitable for home or industrial safety applications.

Future Enhancement

- Integrate **mobile or IoT alerts** for remote monitoring.
- Add **LCD display** to show gas concentration in real-time.
- Improve sensitivity to detect **very low gas concentrations**.
- Implement **automatic ventilation or shut-off systems** for enhanced safety.

CHAPTER 5

REFERENCES

The information and technical knowledge used in this project were referred from the Arduino official website for understanding Arduino UNO hardware and programming concepts. The working principle and specifications of the MQ series gas sensor were referred from standard sensor datasheets and technical documentation available online. Additional guidance for circuit connections, coding logic, and implementation methodology was obtained from various Arduino-based gas leak detection tutorials available on YouTube and other educational websites.

General technical concepts and troubleshooting support were also referred from online electronics forums and web resources related to embedded systems and gas sensing applications.